

Logic Design and Switching Theory

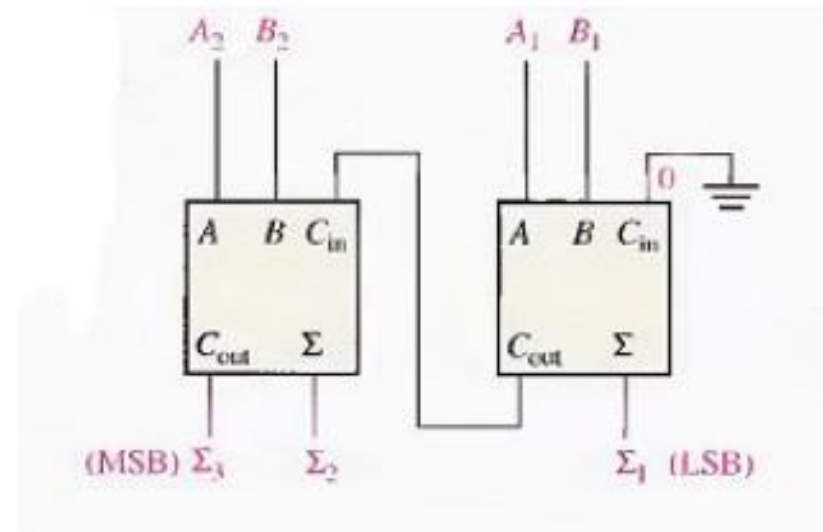
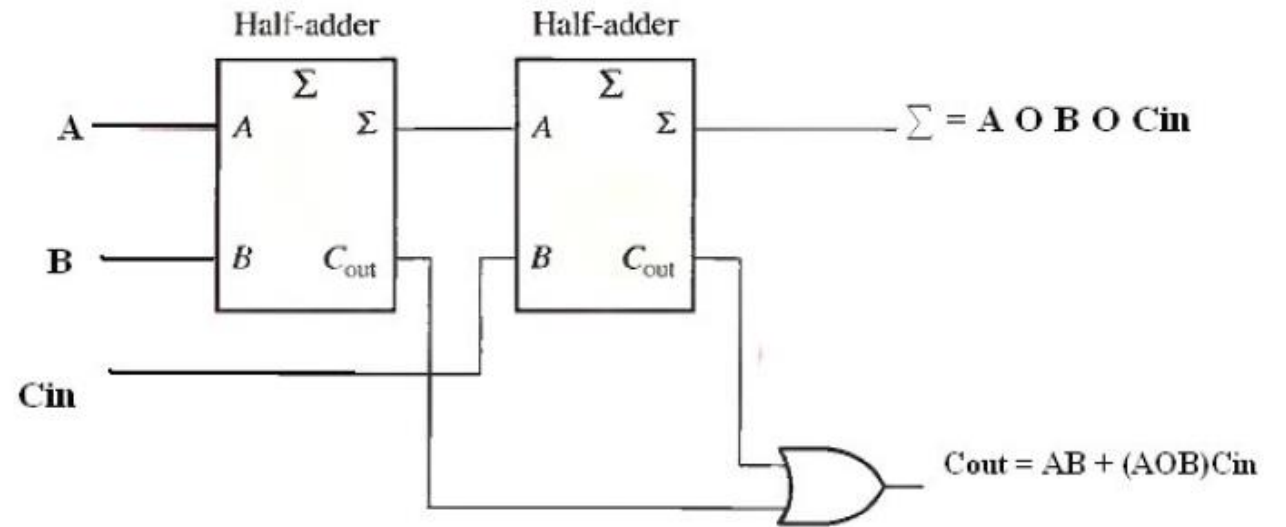
Lecture – 7

Subtitle: Decoders, Encoders, Multiplexers, and
Demultiplexers

by
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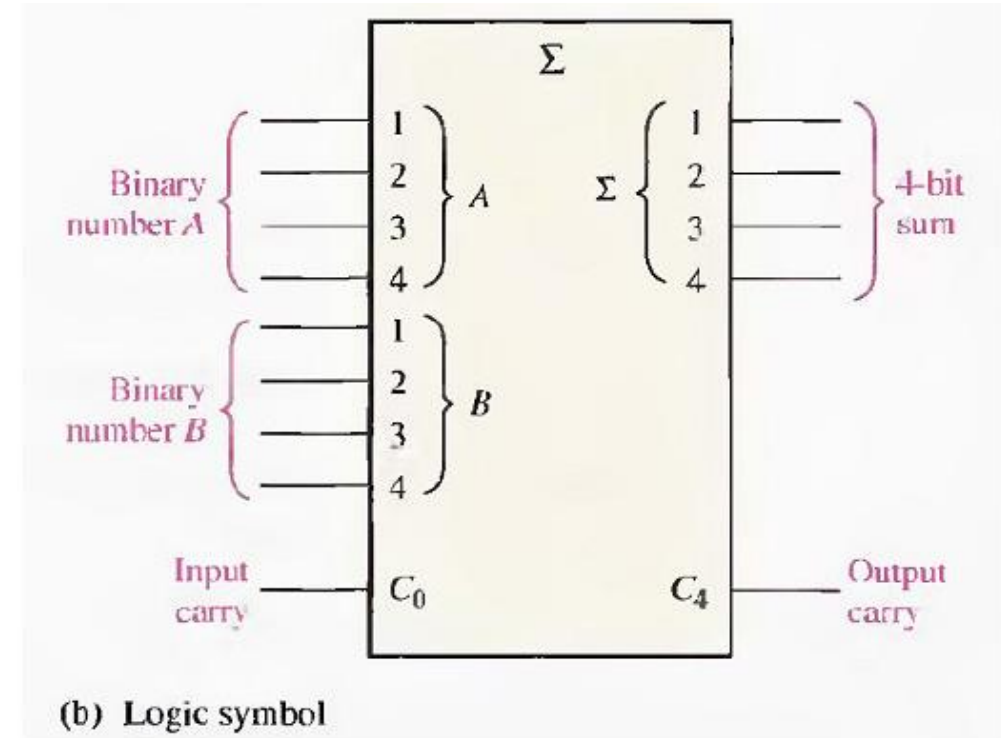
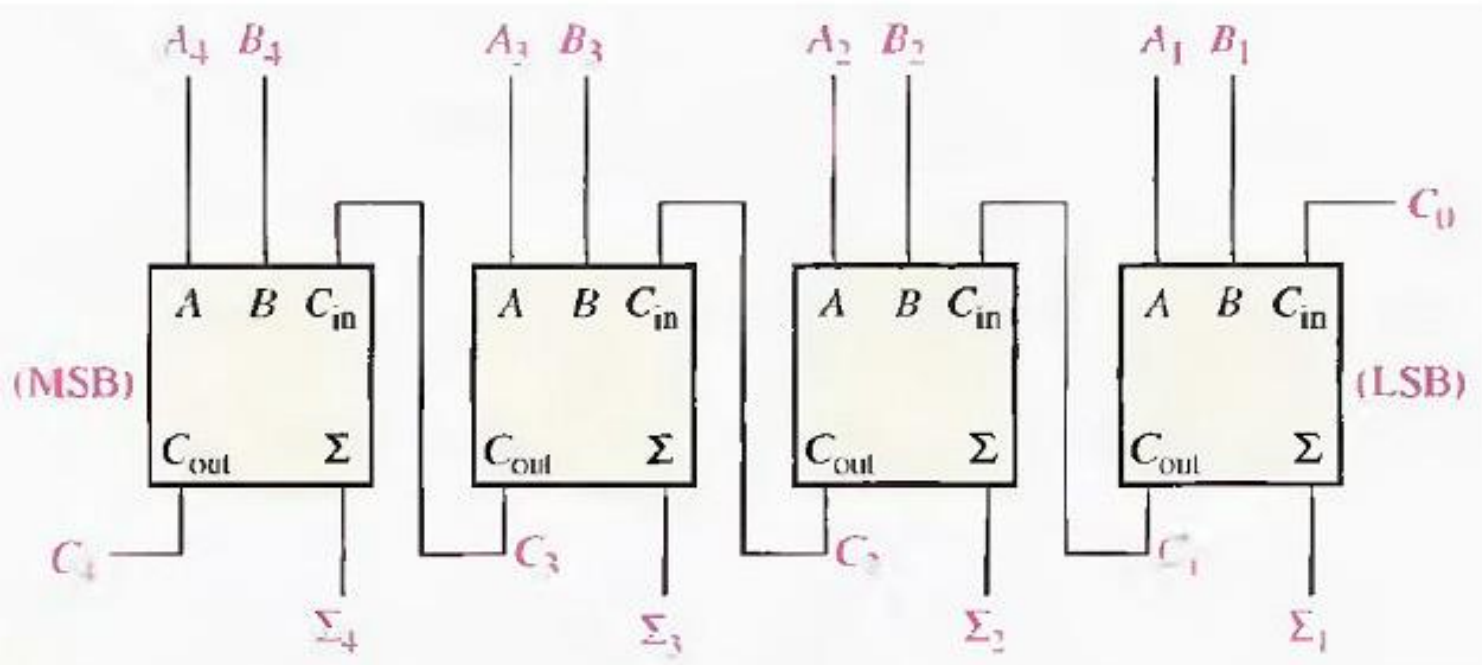
Previous

- Basic Adders
- Parallel adders
- Carry Look Ahead Adder

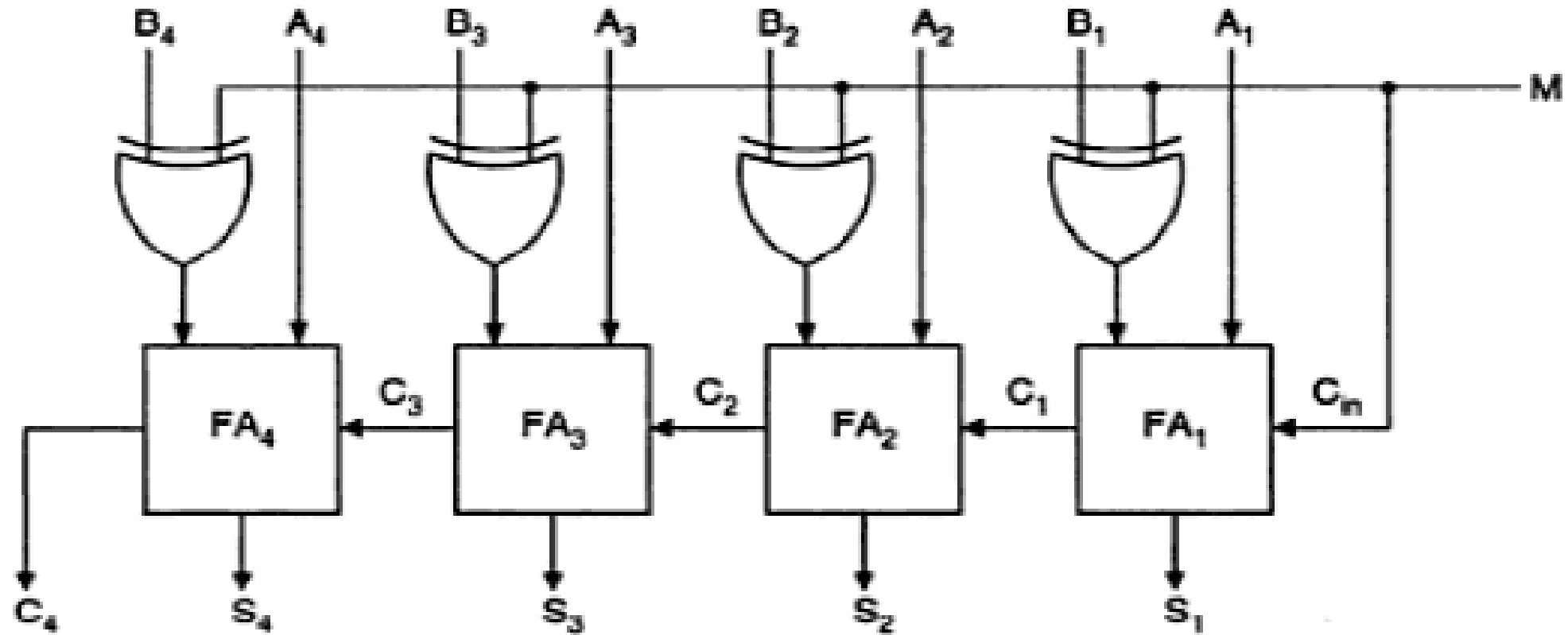


Block diagram of a basic 2-bit parallel adder using two full-adders.

4 Bit Parallel Adders



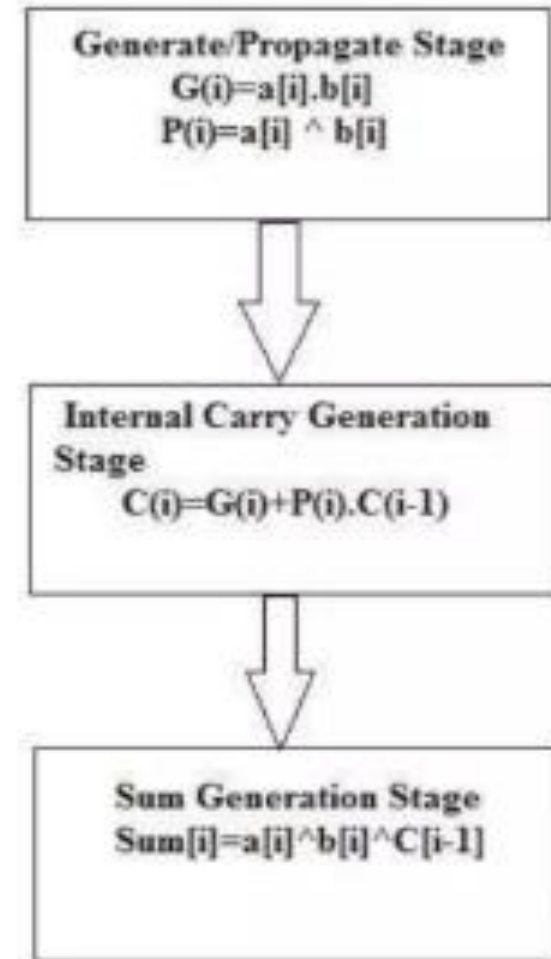
Adder/Subtractor



Logic diagram of a 4-bit binary adder-subtractor.

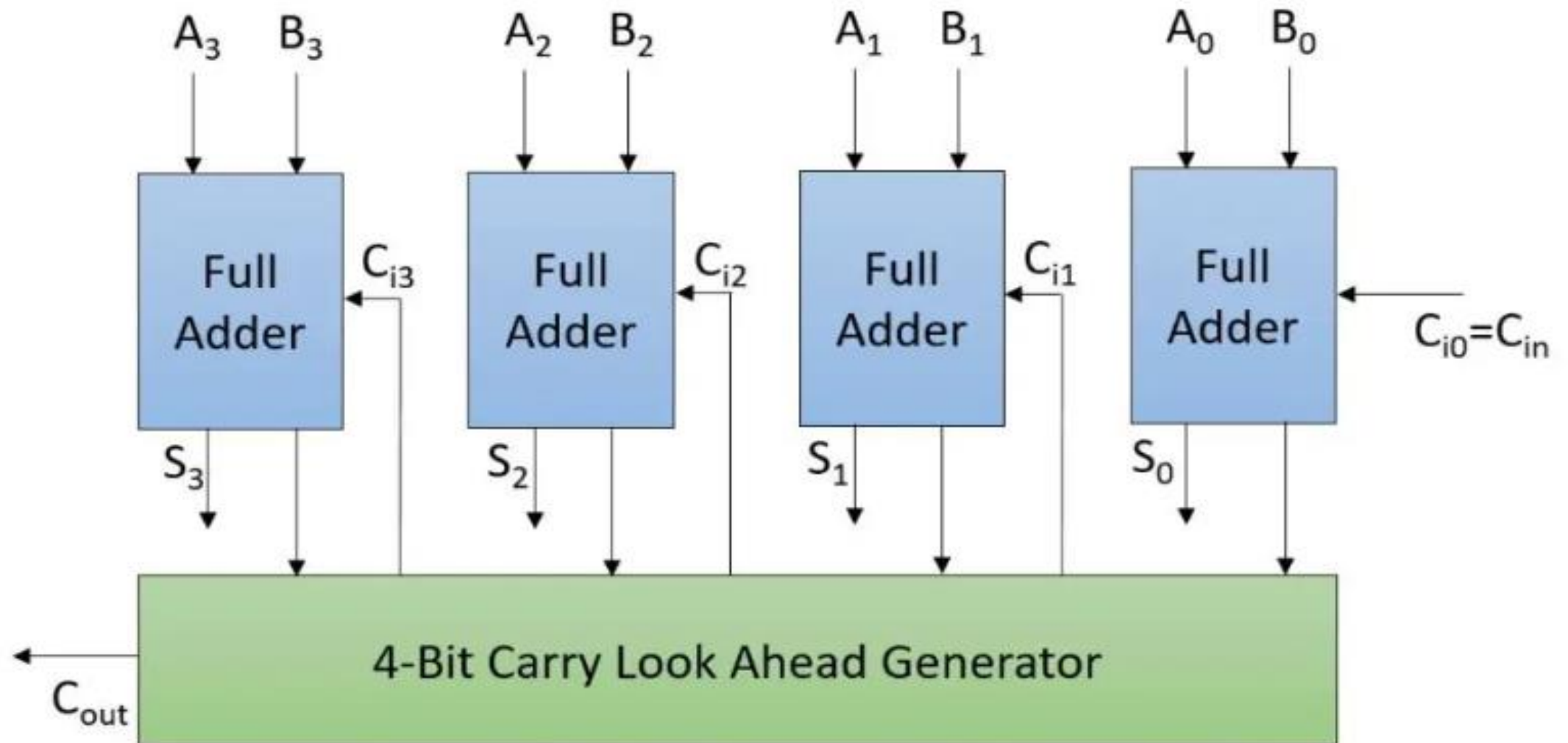
Carry Look-ahead Adder

Carry Look Ahead (CLA) design is based on the principle of looking at lower adder bits of argument and addend if higher orders carry generated. This adder reduces the carry delay by reducing the number of gates through which a carry signal must propagate. As shown in fig. in the generation and propagation stage, the generation values, propagation values are computed. Internal carry generation is calculated in second stage. And in final stage, the sum is calculated. The flow chart of CLA is given in fig. and the architecture of CLA is given in fig.



Flow chart of CLA

Carry Look-ahead Adder



Carry Look-ahead Adder

Disadvantage

Requires complex hardware to generate the carry based on input signals and the complexity grows with an increasing number of bits.

Advantage

It does not need to wait till the previous stage generates the carry. It can generate carry for each full adder simultaneously, hence it is faster as compared to the ripple carry adder.

Functions of Combinational Logic

Decoder, MUX and DMUX

Decoder, MUX and DMUX

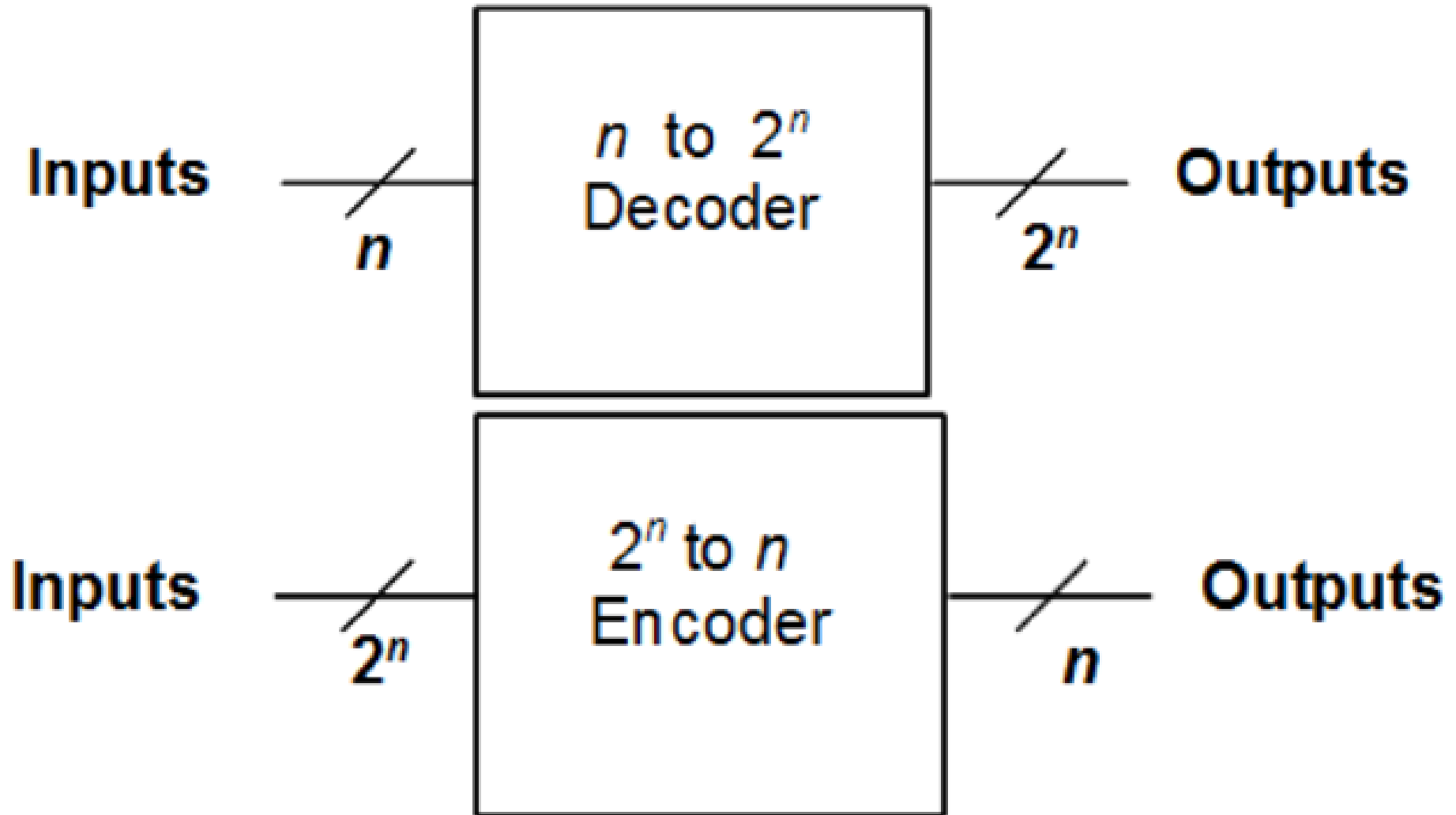
Digital circuits use data selection and distribution methods.

Three essential components:

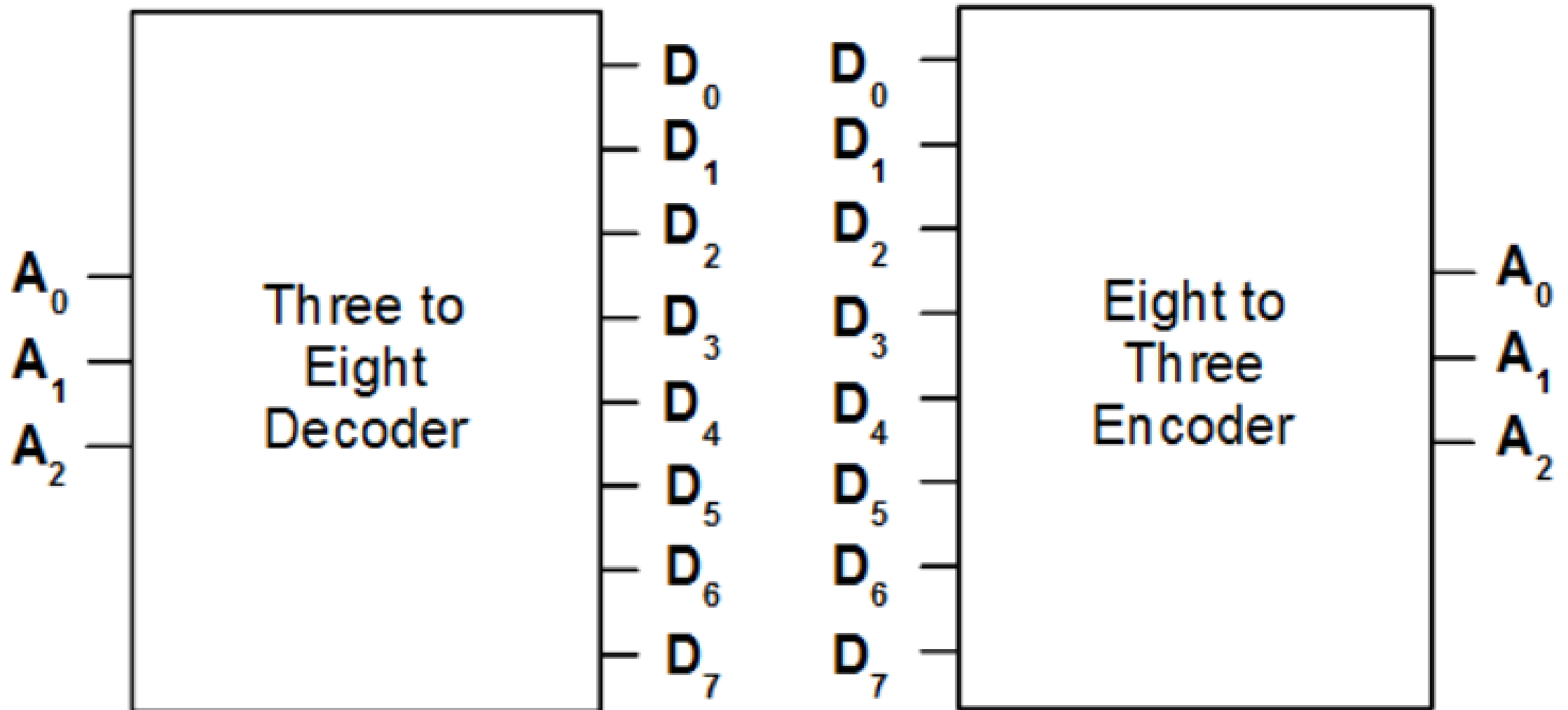
1. **Decoder,**
2. **Multiplexer (MUX), and**
3. **Demultiplexer (DEMUX).**

- A **decoder** is a combinational circuit that converts binary input into a unique active output. Used in memory addressing, instruction decoding, and display systems.
- A **multiplexer (MUX)** is a combinational circuit that selects one input from multiple inputs and forwards it to a single output. Acts as a data selector. Common types: **2-to-1, 4-to-1, 8-to-1 MUX.**
- A **demultiplexer (DEMUX)** is a combinational circuit that takes a single input and routes it to one of many outputs based on control signals. Acts as a data distributor. Common types: **1-to-2, 1-to-4, 1-to-8 DEMUX.**

Inputs and Outputs of Decoder and Encoder



Decoder 3x8 and Encoder 8x3



Decoder

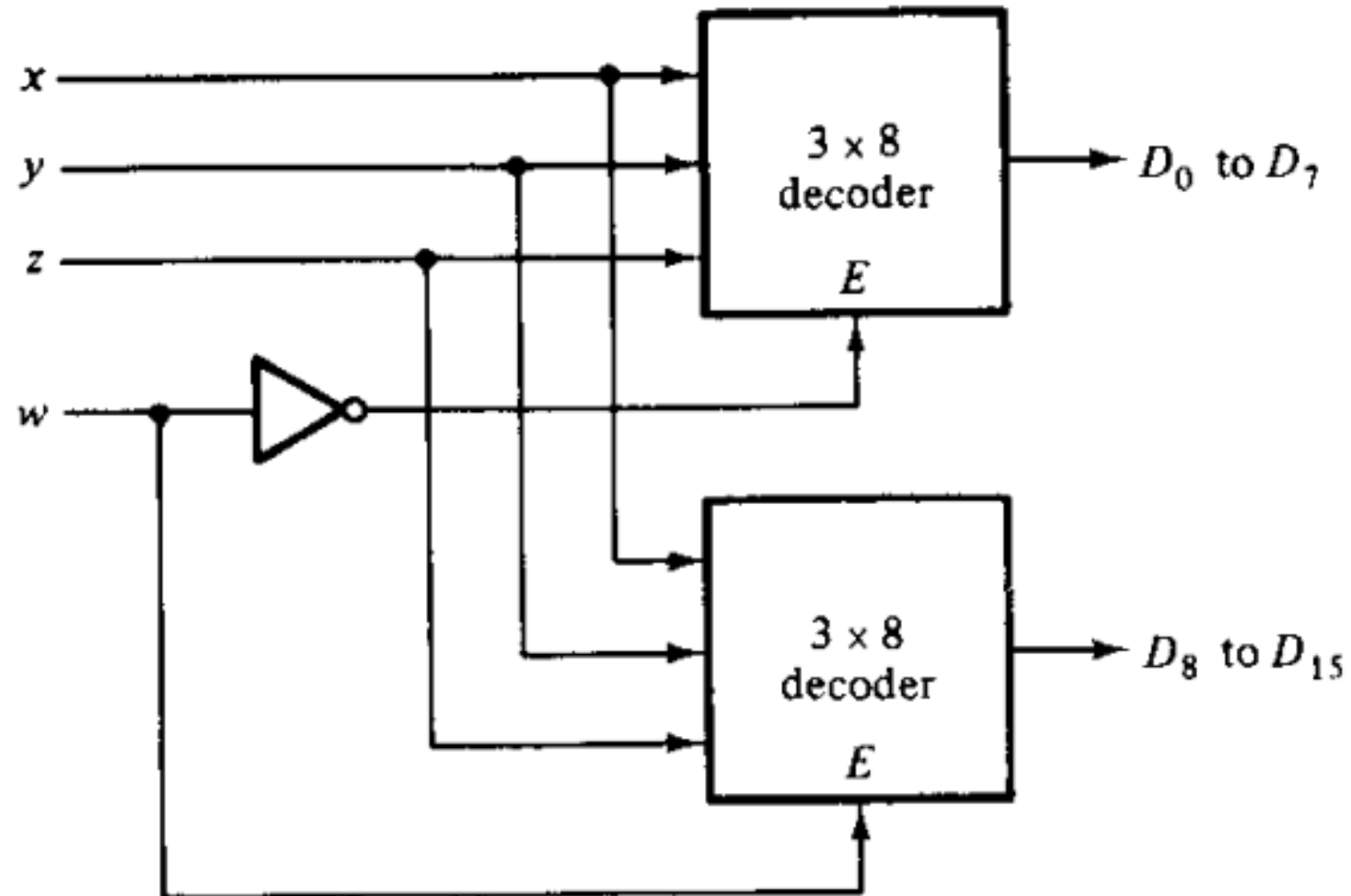
A_2	A_1	A_0	D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0
0	0	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	1	0	0	0	0	0	0	1	0	0
0	1	1	0	0	0	0	1	0	0	0
1	0	0	0	0	0	1	0	0	0	0
1	0	1	0	0	1	0	0	0	0	0
1	1	0	0	1	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0

Decoder Applications

- Decoders are widely used in memory systems of computers where they respond to the address code input from the central processor to activate the memory storage location specified by the address code.
- In every wireless communication, data security is the main concern. The decoders are mainly designed to provide security for data communication by designing standard encryption and decryption algorithms.
- Decoders are used in audio systems to convert analogue audio into digital data.
- Used as a decompressor to convert compressed data like images and videos into decompressed form.
- Decoders use electronic circuits which convert computer instructions into CPU control signals.

Q: Cascade two 3x8 line Decoder to form a 4x16 Decoder

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74138: 3x8 line decoder

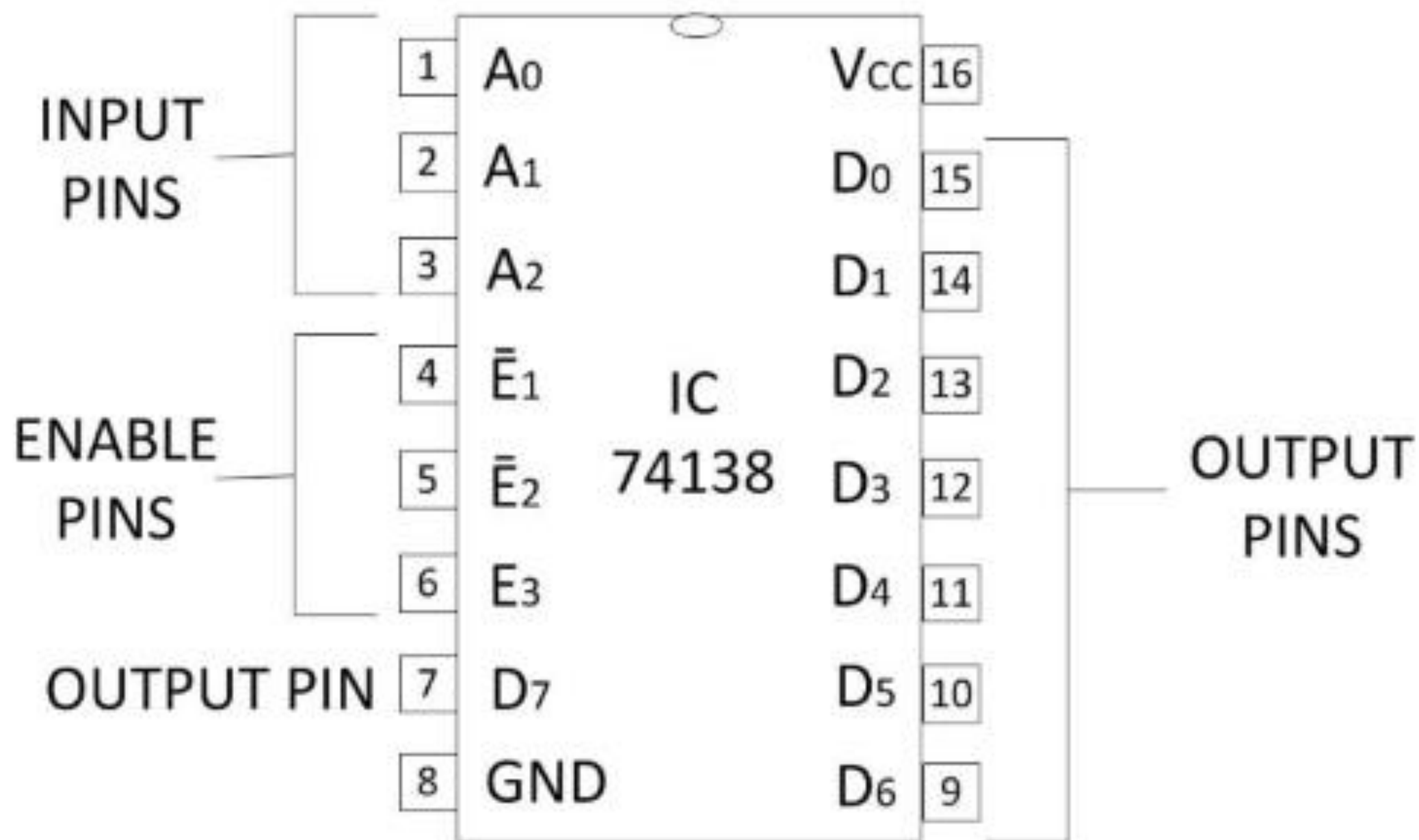
The **74138 (74LS138)** is a **3-to-8 line decoder/demultiplexer**. It takes a **3-bit binary input** and activates one of its **8 outputs** (D0 to 7) while keeping the others LOW. This IC is commonly used in memory addressing, demultiplexing, and other digital logic applications.

The decoder works only when:

- **G1 = 1** (HIGH)
- **G2A = 0** (LOW)
- **G2B = 0** (LOW)

Features of 74138

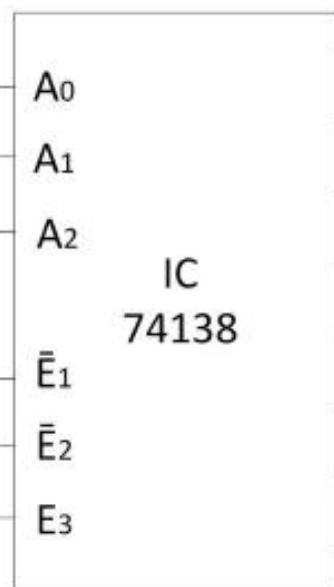
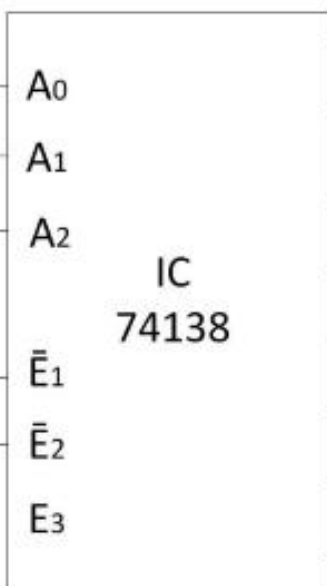
- It is very fast and high speed IC.
- It consumes very low power because it consists of low power Schottky diodes.
- It has a demultiplexing facility.
- It has a very short propagation delay.
- Appropriate operating temperature is from 0°C to 70°C.



Connect two 74138 (3-to-8) decoders to obtain a 4-to-16 decoder. Make a Logic diagram.

BINARY
INPUTS

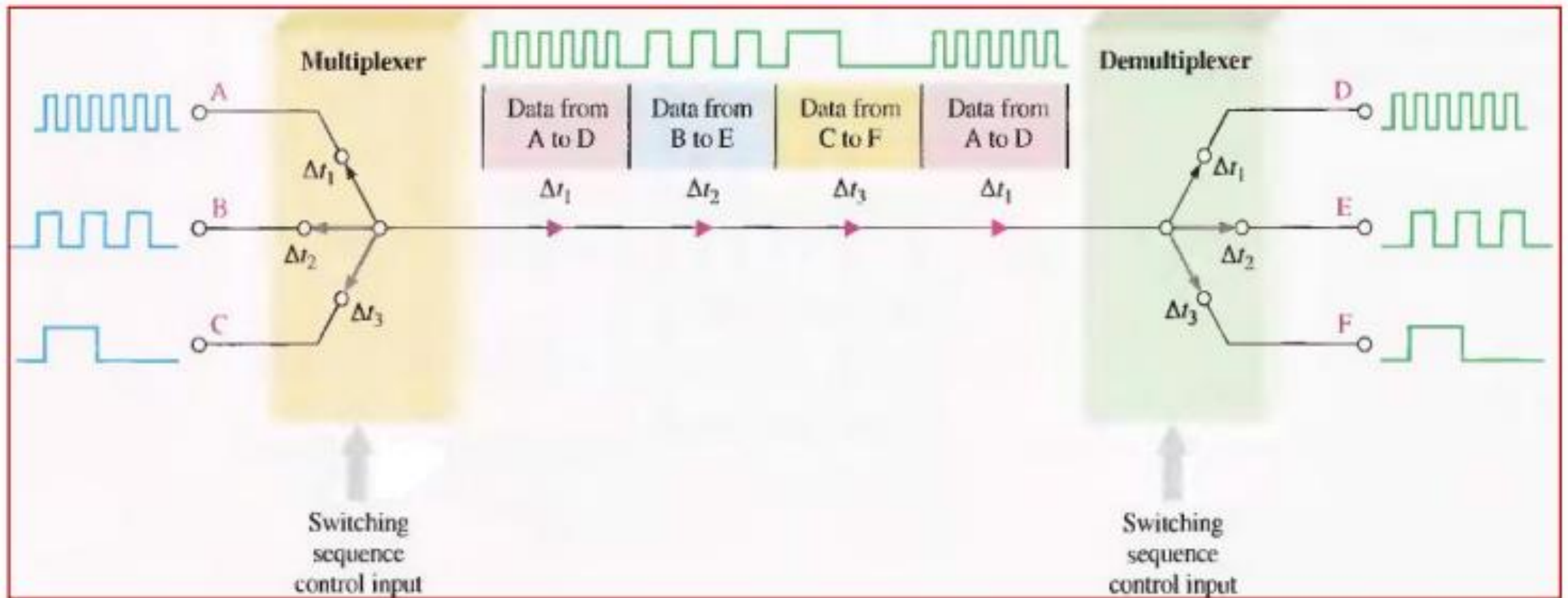
A₀
A₁
A₂
A₃



DECIMAL
OUTPUTS

MUX and DMUX

Multiplexing and De-Multiplexing are used when data from several sources are to be transmitted over one line to a distant location and redistributed to several destinations.



MUX

1. Time Division Multiplexing (TDM)

Concept: Each signal is assigned a specific time slot in a cyclic manner.

Types:

- **Synchronous TDM:** Fixed time slots for each signal, even if no data is present.
- **Asynchronous (Statistical) TDM:** Time slots are dynamically allocated based on data availability.

Example: Digital telephony, where multiple phone calls share the same transmission line.

2. Frequency Division Multiplexing (FDM)

Concept: Each signal is assigned a unique frequency band within the available spectrum.

Working: Signals are modulated at different carrier frequencies and transmitted simultaneously.

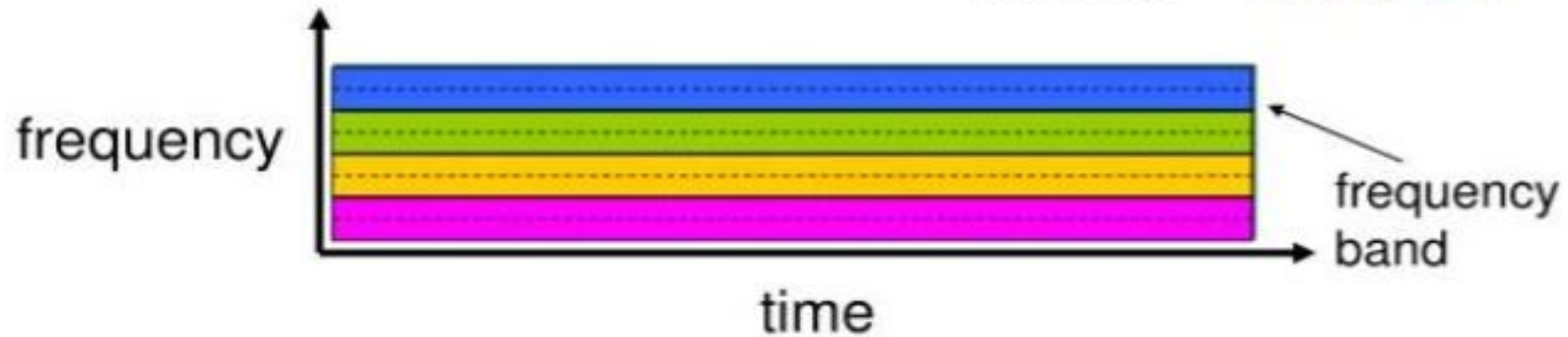
Example: FM/AM radio broadcasting, where different stations operate at different frequencies.

FDM vs TDM

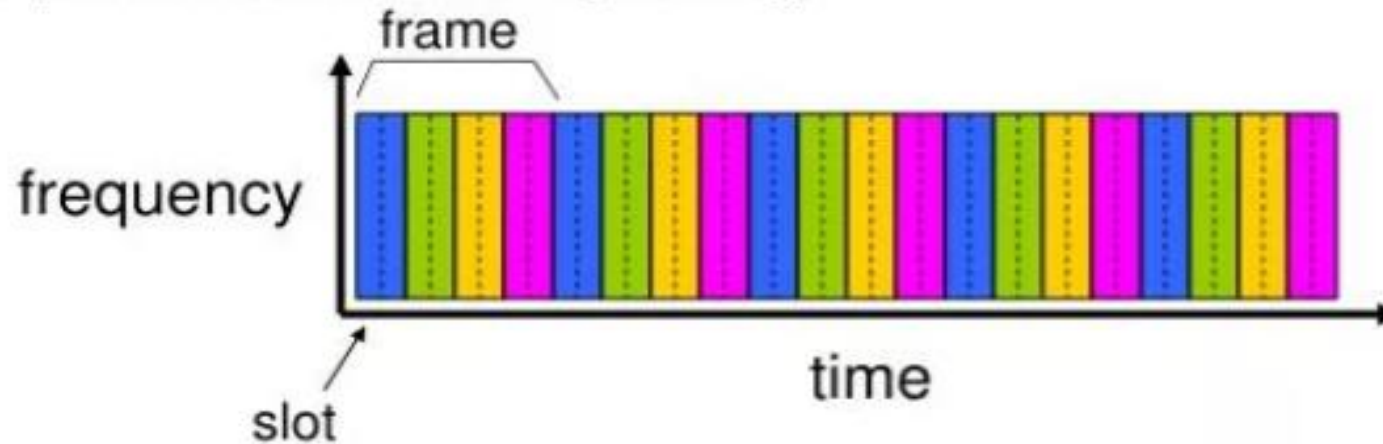
FDM (Frequency division multiplexing)

Example:

4 users

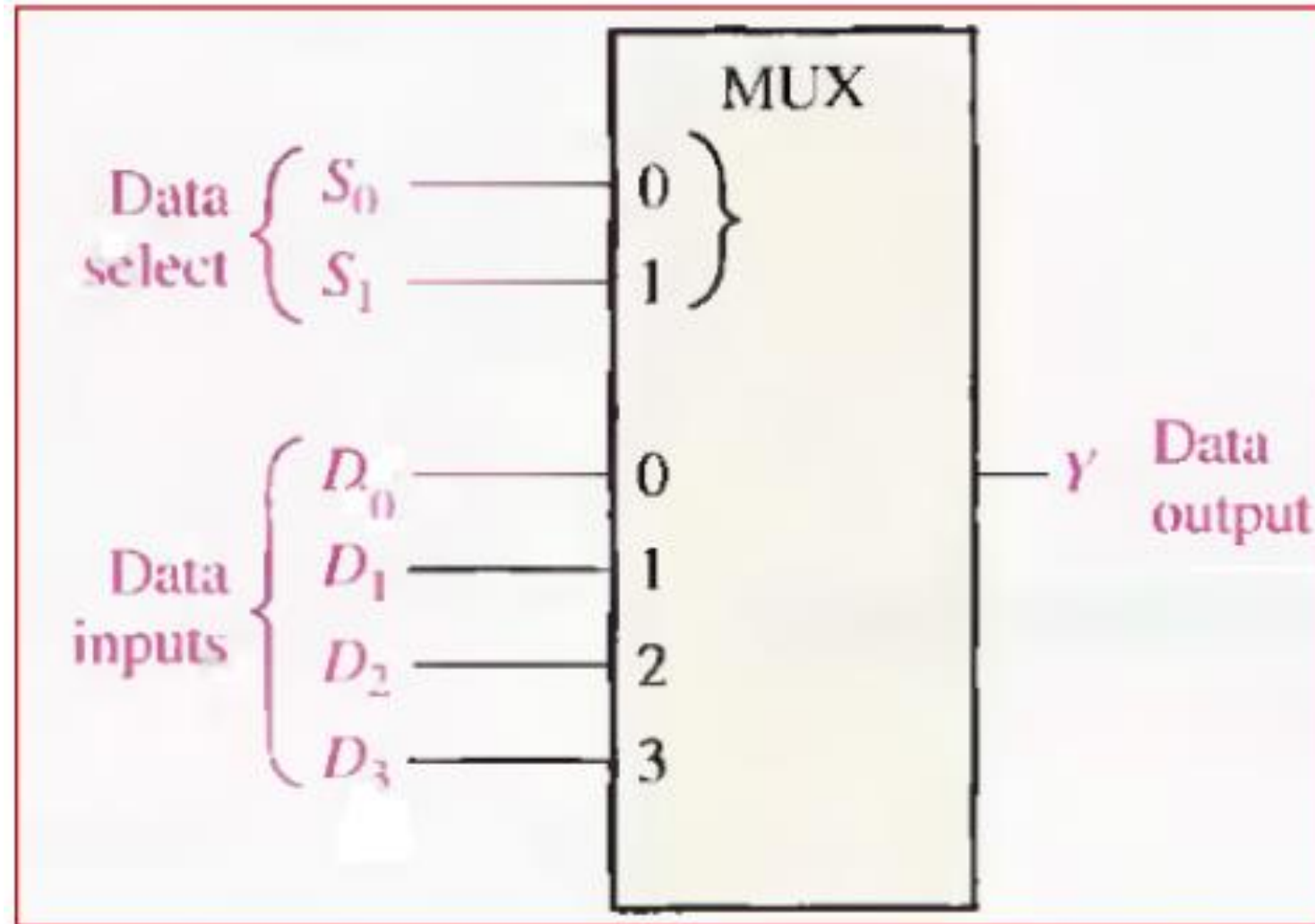


TDM (Time division multiplexing)



Features	FDM	TDM
Basis	Time-based allocation	Frequency-based allocation
Signal Type	Mostly used for digital signals	Mostly used for analog signals
Bandwidth Usage	Full bandwidth shared in time	Bandwidth is divided into multiple frequency bands
Synchronization	Requires precise synchronization	No synchronization required between channels
Crosstalk	No crosstalk (separate time slots)	Possible crosstalk due to adjacent frequency bands
Efficiency	More efficient for bursty traffic	More efficient for continuous signals
Implementation Complexity	Requires timing circuits	Requires bandpass filters and modulators
Example Applications	Digital telephony, ISDN, Ethernet	FM/AM radio, Cable TV, LTE

4 input MUX



4 input MUX

There are two data-select lines because with two select bits. Anyone of the four data-input lines can be selected.

DATA-SELECT INPUTS		INPUT SELECTED
S_1	S_0	
0	0	D_0
0	1	D_1
1	0	D_2
1	1	D_3

4 input MUX

The data output is equal to the state of the selected data input.

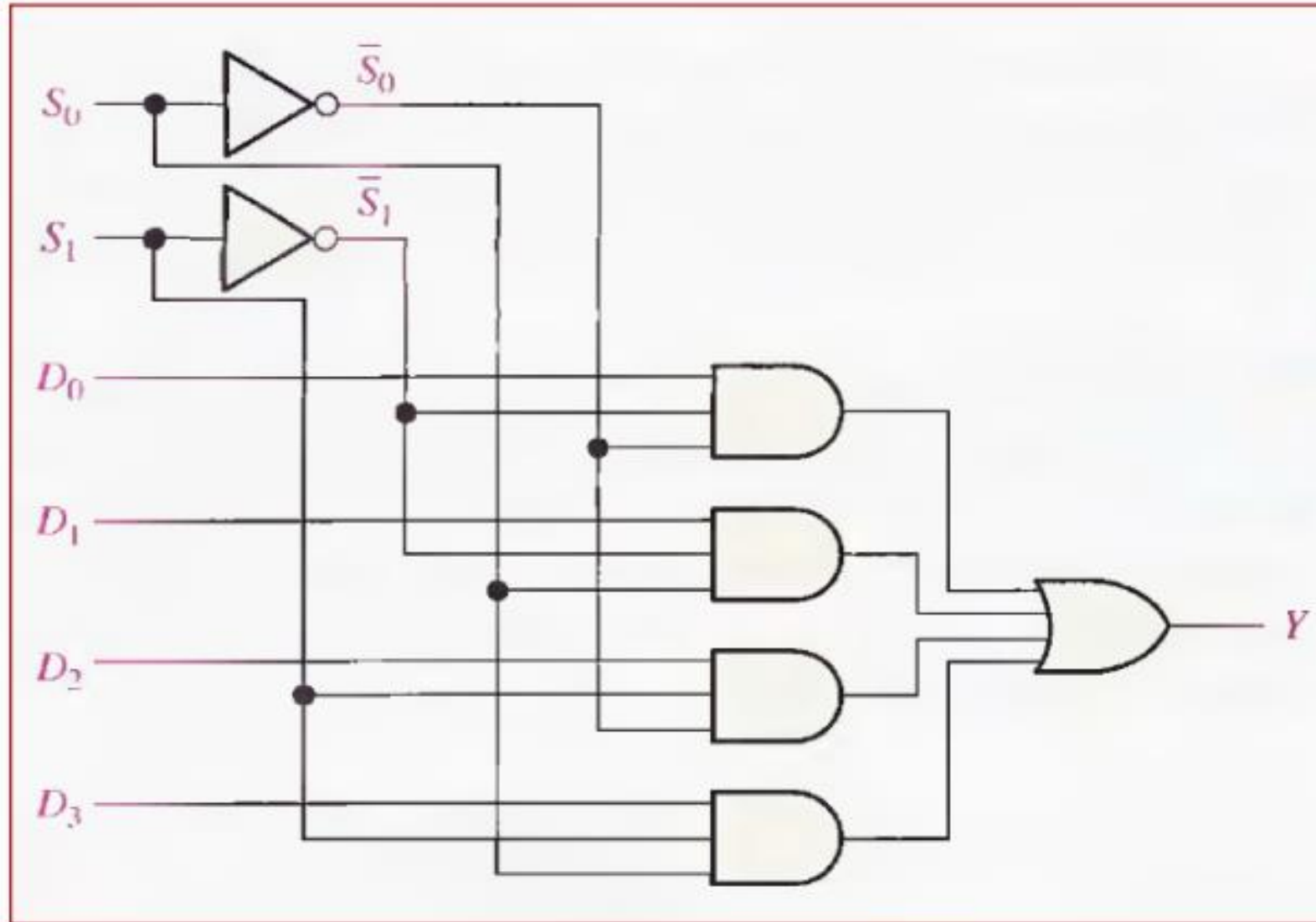
You can therefore, derive a logic expression for the output in terms of the data input and the select inputs.

Because data can be selected from anyone of the input lines, this circuit is also referred to as a **data selector**.

Data output	S_1	S_0	Y
D_0	0	0	$D_0 S'_1 S'_0$
D_1	0	1	$D_1 S'_1 S_0$
D_2	1	0	$D_2 S_1 S'_0$
D_3	1	1	$D_3 S_1 S_0$

$$Y = D_0 S'_1 S'_0 + D_1 S'_1 S_0 + D_2 S_1 S'_0 + D_3 S_1 S_0$$

4 input MUX



Quadruple

The data output is equal to the state of the selected data input.

You can therefore, derive a logic expression for the output in terms of the data input and the select inputs.

Because data can be selected from anyone of the input lines, this circuit is also referred to as a **data selector**.

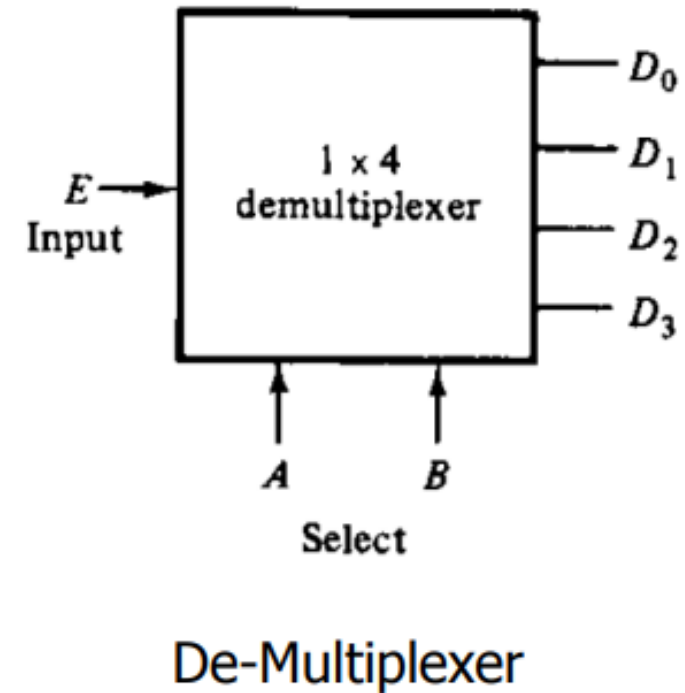
Data output	S_1	S_0	Y
D_0	0	0	$D_0 S'_1 S'_0$
D_1	0	1	$D_1 S'_1 S_0$
D_2	1	0	$D_2 S_1 S'_0$
D_3	1	1	$D_3 S_1 S_0$

$$Y = D_0 S'_1 S'_0 + D_1 S'_1 S_0 + D_2 S_1 S'_0 + D_3 S_1 S_0$$

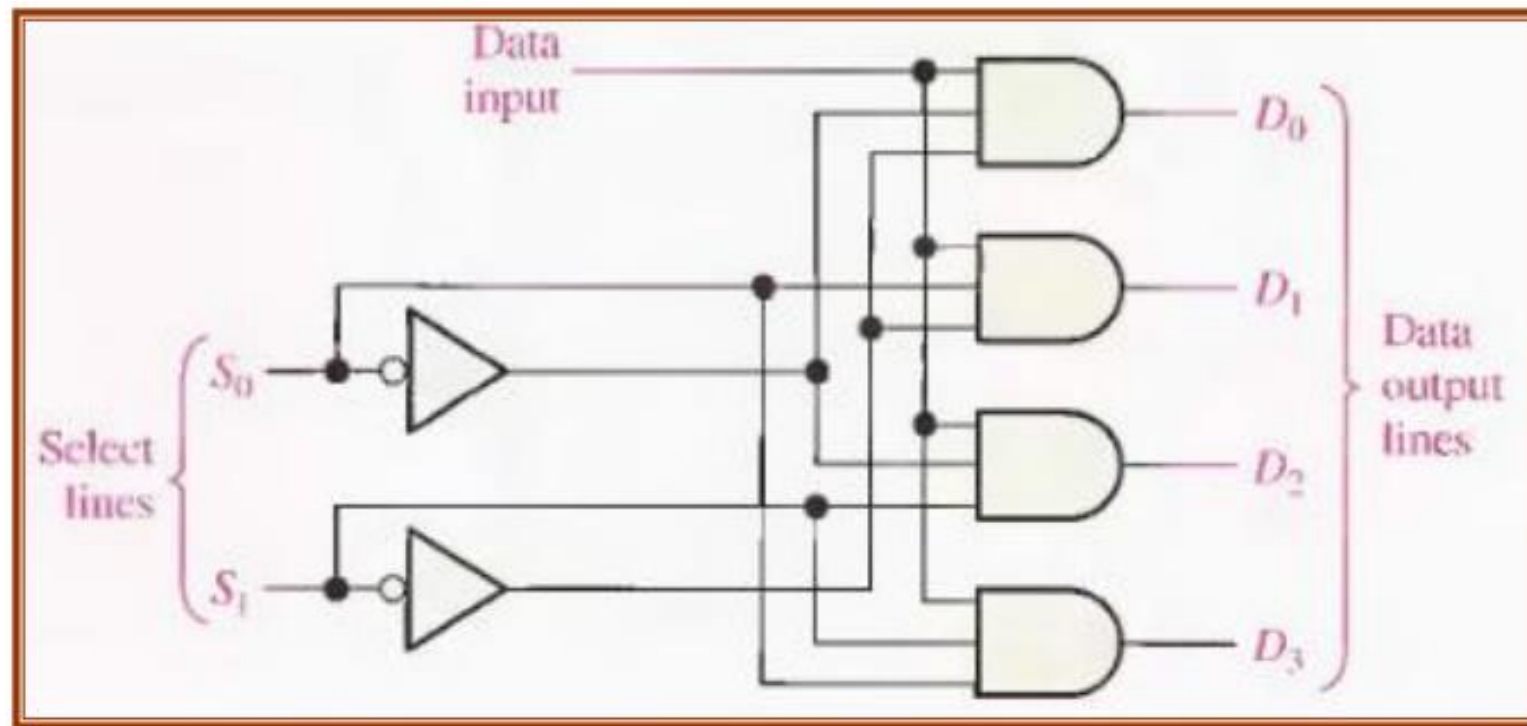
DMUX

Basic functionalities

- A Demultiplexer (DEMUX) basically reverses the multiplexing function.
- It takes digital information from one line and distributes it to a given number of output lines.
- For this reason, the demultiplexer is also known as a data distributor.
- Decoders can also be used as demultiplexers.



1x4 DMUX



E	A	B	D_0	D_1	D_2	D_3
1	X	X	1	1	1	1
0	0	0	0	1	1	1
0	0	1	1	0	1	1
0	1	0	1	1	0	1
0	1	1	1	1	1	0

MUX implementation

If $F(A,B,C) = (1,2,4,5)$

If A and B is selected as Selected input, then

<i>A</i>	<i>B</i>	<i>C</i>	<i>F</i>
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

$$F = C$$

$$F = C'$$

$$F = 1$$

$$F = 0$$

MUX implementation

Implement the following function with a multiplexer:

$$F(A, B, C, D) = \Sigma(0, 1, 3, 4, 8, 9, 15)$$

Applications of MUX and DMUX

MUX Applications:

- Data selection in communication systems
- Signal routing in ALUs
- Efficient use of transmission lines

DEMUX Applications:

- Data demultiplexing in networking
- Address decoding in memory
- Control signal distribution

Summary

- **Decoder translates binary input to unique output.**
- **MUX selects one input among multiple sources.**
- **DEMUX distributes input to multiple outputs.**
- **Essential in digital circuits for data processing and communication.**

Practice Problems

1. Design a 4x16 decoder
2. Cascade two 4x16 decoder to form a 5x32 decoder.
3. Cascade four 3x8 decoders to form a 5x32 decoder
4. Design BCD to Decimal decoder
5. Implement full adder circuit using 2x4 decoder
6. Implement full adder circuit using 3x8 decoder

Thank you!